Neet Mehulkumar Mehta

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ROBOTICS & PERCEPTION ML ENGINEER

Robotics ML engineer with 3+ years of experience developing and deploying autonomous-driving perception systems, specializing in 3D object detection, multi-sensor fusion, and object tracking. Experienced in improving model accuracy and temporal behavior through data-driven experimentation, loss-function optimization, and trajectory analysis. Strong background in production ML deployment, GPU inference optimization, and embedded systems using C++, CUDA, PyTorch, and TensorRT, with experience taking perception components from model development through integration and vehicle-level validation.

EDUCATION

Worcester Polytechnic Institute (WPI) — **Worcester, MA**
Master of Science- Robotics Engineering, GPA- 3.85/4.00 — Dec 2022

Nirma University — **Ahmedabad, India**
Bachelor's in mechanical engineering, GPA- 7.8/10.00 — May 2020

TECHNICAL SKILLS

- **Programming**: C++, Python, CUDA
- **Machine Learning & Perception**: PyTorch, 3D Object Detection, Object Tracking, Multi-Sensor Fusion, BEV Perception, Monocular Depth Estimation
- **Inference & Optimization**: TensorRT, ONNX
- **Robotics & Simulation**: ROS2, CARLA, OpenCV
- **Engineering & Infrastructure**: Docker, Git, CI/CD

RELEVANT WORK EXPERIENCE

Zenseact — **Gothenburg, Sweden**
Deep Learning Consultant — Apr 2026 – Present
C++, CUDA, Python, Pytorch, TensorRT

- Develop and improve 3D object detection and tracking models for autonomous driving, focusing on model training, loss-function optimization, trajectory analysis, and evaluation.
- Improved 3D object detection AP from **0.66 to 0.68** by adjusting training loss weighting to account for objects annotated in ground truth but unobservable with the target sensor configuration.
- Improved evaluation metrics across curved trajectories by addressing underrepresentation of turning vehicles through targeted trajectory oversampling and loss reweighting.
- Implemented a **3D object tracker from scratch** for an experimental detection model, developing the detection-association and tracking pipeline.
- Served as a primary contributor to the development of a vehicle-light state detection component, taking it from proof of concept to deployment in approximately **two months** through team collaboration.
- Trained a vehicle-light classification model achieving **0.83 overall F1 score** for turn indicators, brake lights, and tail lights; developed the image-patch extraction and temporal state-estimation pipeline using EMA and an FSM with hysteresis, and deployed the component using TensorRT and C++.

Aavaran — **Bhuj, India**
Applied Generative Vision Engineer, Self Employed — Mar 2025 – Mar 2026
Pytorch, ComfyUI, Docker, Vastai

Applied generative vision techniques as an extension of my perception and geometry background, focusing on spatial alignment, scale consistency, and deployment efficiency.

- Founded and operated a small furnishing business, gaining hands-on experience in entrepreneurship, product development, and business operations.
- Developed image-conditioned generative vision workflows using pretrained models, LoRA fine-tuning, and spatial conditioning techniques for virtual try-on and interior visualization.
- Optimized generative model inference across GPU environments using quantized models and custom workflows.

TORC Robotics — **Austin, TX**
ML Engineer II — Feb 2023 – Feb 2025
C++, CUDA, ROS2, Python, Pytorch, AWS, TensorRT

- Developed a multi-sensor BEV-based 3D object detection model, extending a BEVFusion baseline with depth-aware camera feature lifting and cross-modal feature fusion.
- Designed and implemented an attention-based spatio-temporal feature alignment module to align LiDAR and camera BEV features, improving detection mAP by **2.5 percentage points** over the baseline on proprietary datasets.
- Optimized model inference through PyTorch-to-ONNX-to-TensorRT conversion and FP16 post-training quantization, reducing end-to-end latency by approximately **50%** on embedded hardware.
- Implemented custom CUDA kernels and TensorRT plugins for performance-critical operations, including BEV pooling, voxelization, non-maximum suppression (NMS), and bounding-box decoding.
- Integrated the optimized perception model into a production inference engine and developed a sensor-agnostic ROS2 perception node supporting configurable camera and LiDAR inputs.
- Supported deployment of the perception stack on a production autonomous truck, following a V-model development process with unit testing, software-in-the-loop (SIL), and hardware-in-the-loop (HIL) validation.
- Built an automated ML lifecycle pipeline using Comet and CI/CD for model training, versioning, ONNX/TensorRT conversion, validation, and deployment.

Research — Sim-to-Real Adaptation and Monocular Depth Estimation

- Developed a proof of concept for geometry-guided unsupervised domain adaptation using synthetic driving data, combining semantic segmentation and monocular depth estimation to address the simulation-to-real domain gap.

TORC Robotics **Blacksburg, VA**

Perception Engineer – Co-Op Jan 2022 – Aug 2022

C++, Python, Pytorch, AWS, TensorRT

- Worked on Data extraction and data postprocessing for deep learning architectures. Established extendable pipeline to generate detailed metrics report for each Deep learning model.
- Developed automated hyperparameter tuning stage in AWS Sagemaker. Used Bayesian search to find optimal hyperparameters.

PROJECTS

Synthetic Data Generation for Self Driving Cars using CARLA simulator Aug 2022 – Present

Python, Pytorch, CARLA

- Built a configurable synthetic-data generation toolkit for autonomous-driving perception, supporting depth, optical flow, semantic segmentation, and 2D/3D object detection.
- Developed a modular multi-sensor simulation pipeline with configurable sensor placement, camera intrinsics/extrinsics, traffic, pedestrians, scene complexity, and capture frequency; generated time-synchronized multimodal annotations to support sensor fusion, domain adaptation, and self-supervised learning.